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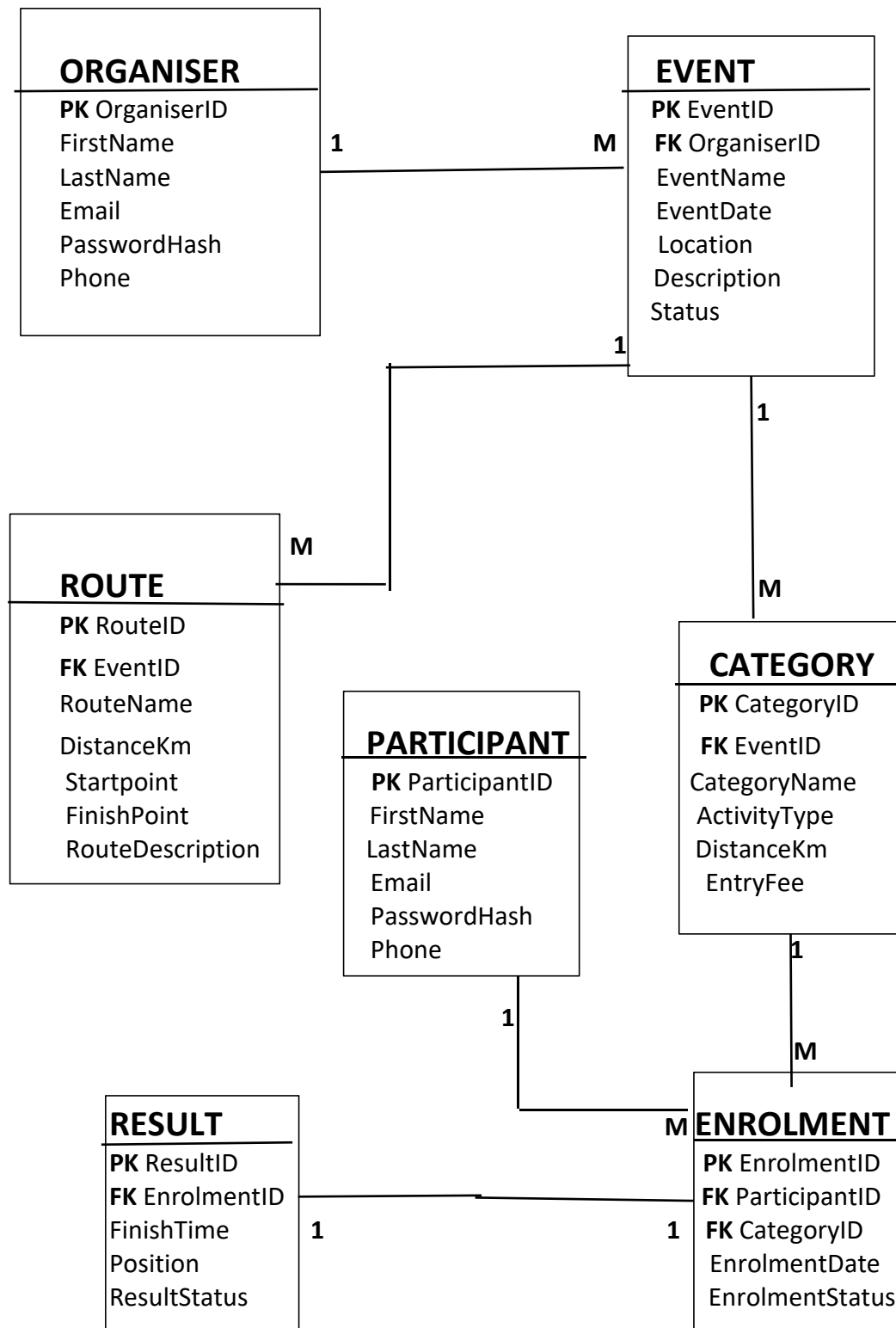
MODULE NAME : Programming 2B

MODULE CODE : PROG6212/w

ASSESSMENT TYPE : POE

PART 1 - SYSTEM PLANNING AND DATABASE

Section A - Entity Relationship Diagram (ERD)



Section B - RaceDay API Endpoint Plan

HTTP METHOD	ROUTE	DESCRIPTION	ROLE REQUIRED	REQUEST BODY	EXPECTED RESPONSE
POST	/api/auth/register	Registers a new user and creates an account with either the organizer or the participant role. Role selection should be validated to prevent unauthorized access to organiser functionality.	None	{firstName, lastName, email, password, role}	201 Created – user registered; 400 Bad Request – invalid data; 409 Conflict – email already exists
POST	/api/auth/login	Authenticates a registered user and establishes their authenticated session and role.	None	{email, password}	200 OK – login successful; 401 Unauthorized – invalid credentials
GET	/api/users/profile	Allows the logged-in user to view their own profile.	Any	None	200 OK – profile returned; 401 Unauthorized – not logged in
PUT	/api/users/profile	Allows the logged-in user to update their own profile information.	Any	{firstName, lastName, email}	200 OK – profile updated; 400 Bad Request – invalid data; 401 Unauthorized – not logged in

GET	/api/events	Returns all available events that users can browse.	None	None	200 OK – list of events
GET	/api/events/{id}	Returns details for a specific event.	None	None	200 OK – event details; 404 Not Found – event does not exist
POST	/api/events	Creates a new road,running, walking or cycling event.	Organiser	{name, description, date, location, distance,eventType}	201 Created – event created; 400 Bad Request – invalid data; 401 Unauthorized – not logged in; 403 Forbidden – wrong role
PUT	/api/events/{id}	Updates an existing event.	Organiser	{name, description, date, location, distance, eventType}	200 OK – event updated; 404 Not Found – event does not exist; 403 Forbidden – wrong role
DELETE	/api/events/{id}	Deletes an existing event.	Organiser	None	204 No Content – event deleted; 404 Not Found – event does not exist; 403 Forbidden – wrong role
GET	/api/events/{eventid}/categories	Returns the categories available for a specific event.	None	None	200 OK – list of categories; 404 Not Found – event does not exist
POST	/api/events/{eventid}/categories	Creates an age or distance category for an event.	Organiser	{name, description}	201 Created – category created; 400 Bad Request – invalid data; 403 Forbidden – wrong role
PUT	/api/categories/{id}	Updates an existing	Organiser	{name, description}	200 OK – category updated; 404

		event category.			Not Found – category does not exist; 403 Forbidden – wrong role
DELETE	/api/categories/{id}	Deletes an existing event category.	Organiser	None	204 No Content – category deleted; 404 Not Found – category does not exist; 403 Forbidden – wrong role
POST	/api/events/{eventid}/enrolments	Allows a participant to enter an event by selecting an available category.	Participant	{categoryID}	201 Created – enrolment recorded; 400 Bad Request – invalid category; 401 Unauthorized – not logged in; 403 Forbidden – wrong role; 409 Conflict – already enrolled
GET	/api/enrolments/my	Returns the logged in Participants own event enrolments.	Participant	None	200 OK – participant's enrolments; 401 Unauthorized – not logged in; 403 Forbidden – wrong role
GET	/api/events/{eventid}/enrolments	Returns all participant enrolments for an event managed by the organizer.	Organiser	None	200 OK – list of enrolments; 404 Not Found – event does not exist; 403 Forbidden – wrong role
POST	/api/enrolments/{enrolmentid}/result	Records a participants finishing time and finishing position after an event.	Organiser	{finishTime, finishPosition}	201 Created – result recorded; 400 Bad Request – invalid data; 404 Not Found – enrolment does not exist; 403 Forbidden – wrong role
GET	/api/results/my	Returns the logged-			200 OK – participant's results;

		in participants personal race result.	Participant	None	401 Unauthorized – not logged in; 403 Forbidden – wrong role
GET	/api/events/{eventid}/results	Returns the results recorded for participants in an organizers event.	Organiser	None	200 OK – event results; 404 Not Found – event does not exist; 403 Forbidden – wrong role

Section C - SQL Database Script

SQL CODE

```
/* ===== RACEDAY DATABASE – SECTION C SQL
SCRIPT ===== */
-- Create Database
```

```
USE master;GO
IF DB_ID('RaceDay') IS NOT NULL BEGIN
    ALTER DATABASE RaceDay SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
    DROP DATABASE RaceDay;END;GO
CREATE DATABASE RaceDay;GO
USE RaceDay;GO
```

```
/* ===== 1. ORGANISER TABLE
===== */
```

```
CREATE TABLE Organiser(
```

```
OrganiserID INT IDENTITY(1,1) PRIMARY KEY,  
FirstName VARCHAR(50) NOT NULL,  
LastName VARCHAR(50) NOT NULL,  
Email VARCHAR(100) NOT NULL UNIQUE,  
Password VARCHAR(255) NOT NULL,  
Phone VARCHAR(20) NOT NULL UNIQUE);GO
```

```
/* =====  
===== */
```

2. PARTICIPANT TABLE

```
CREATE TABLE Participant(  
ParticipantID INT IDENTITY(1,1) PRIMARY KEY,  
FirstName VARCHAR(50) NOT NULL,  
LastName VARCHAR(50) NOT NULL,  
Email VARCHAR(100) NOT NULL UNIQUE,  
Password VARCHAR(255) NOT NULL,  
Phone VARCHAR(20) NOT NULL UNIQUE);GO
```

```
/* =====  
===== */
```

3. EVENT TABLE

```
CREATE TABLE Event(  
EventID INT IDENTITY(1,1) PRIMARY KEY,  
OrganiserID INT NOT NULL,  
EventName VARCHAR(100) NOT NULL,  
EventDate DATE NOT NULL,  
Location VARCHAR(150) NOT NULL,  
Description VARCHAR(500),  
Status VARCHAR(20) NOT NULL DEFAULT 'Upcoming',
```

```

CONSTRAINT FK_Event_Organiser
    FOREIGN KEY (OrganiserID)
    REFERENCES Organiser(OrganiserID),

CONSTRAINT UQ_Event_Name_Date
    UNIQUE (EventName, EventDate));GO

```

```

/* =====
===== */

```

4. ROUTE TABLE

```

CREATE TABLE Route(
    RouteID INT IDENTITY(1,1) PRIMARY KEY,
    EventID INT NOT NULL,
    RouteName VARCHAR(100) NOT NULL,
    DistanceKM DECIMAL(6,2) NOT NULL,
    StartPoint VARCHAR(150) NOT NULL,
    FinishPoint VARCHAR(150) NOT NULL,
    RouteDescription VARCHAR(500),

    CONSTRAINT FK_Route_Event
        FOREIGN KEY (EventID)
        REFERENCES Event(EventID),

    CONSTRAINT UQ_Route_Event_Name
        UNIQUE (EventID, RouteName));GO

```

```

/* =====
===== */

```

5. CATEGORY TABLE


```

CREATE TABLE Category(
    CategoryID INT IDENTITY(1,1) PRIMARY KEY,
    EventID INT NOT NULL,
    CategoryName VARCHAR(100) NOT NULL,
    ActivityType VARCHAR(50) NOT NULL,
    DistanceKM DECIMAL(6,2) NOT NULL,
    EntryFee DECIMAL(10,2) NOT NULL DEFAULT 0.00,

    CONSTRAINT FK_Category_Event
        FOREIGN KEY (EventID)
        REFERENCES Event(EventID),

    CONSTRAINT UQ_Category_Event_Name
        UNIQUE (EventID, CategoryName));GO

```

```

/* =====
===== */

```

6. ENROLMENT TABLE

```

CREATE TABLE Enrolment(
    EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID INT NOT NULL,
    CategoryID INT NOT NULL,
    EnrolmentDate DATE NOT NULL DEFAULT GETDATE(),
    EnrolmentStatus VARCHAR(20) NOT NULL DEFAULT 'Active',

    CONSTRAINT FK_Enrolment_Participant
        FOREIGN KEY (ParticipantID)

```

```

REFERENCES Participant(ParticipantID),

CONSTRAINT FK_Enrolment_Category
    FOREIGN KEY (CategoryID)
    REFERENCES Category(CategoryID),

CONSTRAINT UQ_Enrolment_Participant_Category
    UNIQUE (ParticipantID, CategoryID));GO

```

```

/* ===== 7. RESULT TABLE
===== */

```

```

CREATE TABLE Result(
    ResultID INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID INT NOT NULL UNIQUE,
    FinishTime TIME,
    Position INT,
    ResultStatus VARCHAR(20) NOT NULL DEFAULT 'Pending',

    CONSTRAINT FK_Result_Enrolment
        FOREIGN KEY (EnrolmentID)
        REFERENCES Enrolment(EnrolmentID));GO

```

```

/* ===== INSERT SAMPLE ORGANISERS
===== */

```

```

INSERT INTO Organisier (FirstName, LastName, Email, Password, Phone)VALUES ('Thabo', 'Mokoena',
'thabo.mokoena@example.com', 'Pass1234', '0712345678'), ('Nomsa', 'Dlamini',
'nomsa.dlamini@example.com', 'Pass5678', '0723456789');GO

```

```

/* ===== INSERT SAMPLE PARTICIPANTS
===== */
INSERT INTO Participant (FirstName, LastName, Email, Password, Phone)VALUES ('Sipho', 'Nkosi',
'sipho.nkosi@example.com', 'Part1234', '0734567890'), ('Lerato', 'Molefe',
'lerato.molefe@example.com', 'Part5678', '0745678901');GO

/* ===== INSERT 3 EVENTS
===== */

INSERT INTO Event (OrganiserID, EventName, EventDate, Location, Description, Status)VALUES (1,
'Soweto Community Run', '2026-10-10', 'Soweto, Johannesburg',
'A community road running event.', 'Upcoming'),
(1, 'Thembisa Fun Run', '2026-11-15', 'Thembisa, Gauteng',
'A family friendly running event.', 'Upcoming'),
(2, 'Ekurhuleni Road Race', '2026-12-05', 'Ekurhuleni, Gauteng',
'A competitive road running event.', 'Upcoming');GO

/* ===== INSERT ROUTES FOR EACH EVENT
===== */

INSERT INTO Route (EventID, RouteName, DistanceKM, StartPoint, FinishPoint,
RouteDescription)VALUES (1, 'Soweto 10KM Route', 10.00, 'Walter Sisulu Square', 'Orlando
Stadium',
'A 10 kilometre community route through Soweto. '),
(1, 'Soweto 5KM Route', 5.00, 'Walter Sisulu Square', 'Orlando Stadium',
'A shorter route suitable for recreational runners. '),
(2, 'Thembisa 10KM Route', 10.00, 'Thembisa Stadium', 'Thembisa Stadium',
'A 10 kilometre route around Thembisa. '),

```

```

(2, 'Thembisa 5KM Route', 5.00, 'Thembisa Stadium', 'Thembisa Stadium',
'A 5 kilometre fun run route. '),
(3, 'Ekurhuleni 21KM Route', 21.10, 'Ekurhuleni Stadium', 'Ekurhuleni Stadium',
'A half marathon route through Ekurhuleni. '),
(3, 'Ekurhuleni 10KM Route', 10.00, 'Ekurhuleni Stadium', 'Ekurhuleni Stadium',
'A 10 kilometre competitive route. ');GO

```

```

/* ===== INSERT CATEGORIES FOR EACH EVENT
===== */

```

```

INSERT INTO Category (EventID, CategoryName, ActivityType, DistanceKM, EntryFee)VALUES (1,
'10KM Open', 'Running', 10.00, 100.00), (1, '5KM Fun Run', 'Running', 5.00, 50.00),
(2, '10KM Open', 'Running', 10.00, 80.00), (2, '5KM Family Run', 'Running', 5.00, 40.00),
(3, 'Half Marathon', 'Running', 21.10, 150.00), (3, '10KM Race', 'Running', 10.00,
100.00);GO

```

```

/* ===== INSERT SAMPLE ENROLMENTS
===== */

```

```

INSERT INTO Enrolment (ParticipantID, CategoryID, EnrolmentDate, EnrolmentStatus)VALUES (1, 1,
'2026-09-01', 'Active'), (2, 2, '2026-09-02', 'Active'), (1, 3, '2026-09-03', 'Active'),
(2, 5, '2026-09-04', 'Active');GO

```

```

/* ===== INSERT SAMPLE RESULTS
===== */

```

```

INSERT INTO Result (EnrolmentID, FinishTime, Position, ResultStatus)VALUES (1, '00:52:30', 3,
'Completed'), (2, '00:31:45', 1, 'Completed'), (3, '00:55:10', 5, 'Completed'), (4,
'01:48:20', 2, 'Completed');GO

```

```

/* ===== TEST THE DATABASE
===== */
SELECT * FROM Organiser;SELECT * FROM Participant;SELECT * FROM Event;SELECT * FROM Route;SELECT *
FROM Category;SELECT * FROM Enrolment;SELECT * FROM Result;GO

```

SQL BEING EXECUTED

	OrganiserID	FirstName	LastName	Email	Password	Phone	
1	1	Thabo	Mokoena	thabo.mokoena@example.com	Pass1234	0712345678	
2	2	Nomsa	Dlamini	nomsa.dlamini@example.com	Pass5678	0723456789	

	ParticipantID	FirstName	LastName	Email	Password	Phone	
1	1	Sipho	Nkosi	sipho.nkosi@example.com	Part1234	0734567890	
2	2	Lerato	Molefe	lerato.molefe@example.com	Part5678	0745678901	

	EventID	OrganiserID	EventName	EventDate	Location	Description	Status	
1	1	1	Soweto Community Run	2026-10-10	Soweto, Johannesburg	A community road running event.	Upcoming	
2	2	1	Thembisa Fun Run	2026-11-15	Thembisa, Gauteng	A family friendly running event.	Upcoming	
3	3	2	Ekurhuleni Road Race	2026-12-05	Ekurhuleni, Gauteng	A competitive road running eve...	Upcoming	

	RouteID	EventID	RouteName	DistanceKM	StartPoint	FinishPoint	RouteDescription	
1	1	1	Soweto 10KM Route	10.00	Walter Sisulu Square	Orlando Stadium	A 10 kilometre community route through Soweto.	
2	2	1	Soweto 5KM Route	5.00	Walter Sisulu Square	Orlando Stadium	A shorter route suitable for recreational runners.	
3	3	2	Thembisa 10KM R...	10.00	Thembisa Stadium	Thembisa Sta...	A 10 kilometre route around Thembisa.	
4	4	2	Thembisa 5KM Ro...	5.00	Thembisa Stadium	Thembisa Sta...	A 5 kilometre fun run route.	
5	5	3	Ekurhuleni 21KM ...	21.10	Ekurhuleni Stadium	Ekurhuleni Sta...	A half marathon route through Ekurhuleni.	
6	6	3	Ekurhuleni 10KM ...	10.00	Ekurhuleni Stadium	Ekurhuleni Sta...	A 10 kilometre competitive route.	

	CategoryID	EventID	CategoryName	ActivityType	DistanceKM	EntryFee	
1	1	1	10KM Open	Running	10.00	100.00	
2	2	1	5KM Fun Run	Running	5.00	50.00	
3	3	2	10KM Open	Running	10.00	80.00	
4	4	2	5KM Family ...	Running	5.00	40.00	
5	5	3	Half Marathon	Running	21.10	150.00	
6	6	3	10KM Race	Running	10.00	100.00	

	EnrolmentID	ParticipantID	CategoryID	EnrolmentDate	EnrolmentStatus	
1	1	1	1	2026-09-01	Active	
2	2	2	2	2026-09-02	Active	
3	3	1	3	2026-09-03	Active	
4	4	2	5	2026-09-04	Active	
	ResultID	EnrolmentID	FinishTime	Position	ResultStatus	
1	1	1	00:52:30.0000000	3	Completed	
2	2	2	00:31:45.0000000	1	Completed	
3	3	3	00:55:10.0000000	5	Completed	
4	4	4	01:48:20.0000000	2	Completed	